

Radon Monitoring 4.6.0 - User Manual

Local monitoring for GQ RadonScan devices

Version 4.6.0 · 25 July 2026

1. Purpose and limitations

Radon Monitoring reads completed hourly values from compatible GQ RadonScan devices using read-only communication. It stores raw and converted values locally, publishes Home Assistant entities, analyses time series and creates scientifically structured reports.

The app is a community implementation. It does not replace certified calibration or an officially recognised long-term measurement. Short-term values, statistical relationships and reports must not be used alone for medical, legal or building decisions.

2. Installation and upgrade

Connect the detector by USB, provide MQTT and start the app. Leave “serial_port” empty for automatic detection; prefer a persistent /dev/serial/by-id/ path for fixed assignments.

Download a complete backup before major upgrades or deletion. The slug, database path and MQTT identities are retained. After upgrading, the sidebar must show version 4.6.0.

3. Overview

The Overview shows the last completed hour, data age, site, raw CPH, stored hours, device/MQTT state and the 24-hour, 7-day and 30-day values.

If the period or minimum coverage is incomplete, the tile explains what is missing instead of displaying a misleading number.

4. Analysis

Core statistics include mean, median, 95th percentile, trend, threshold duration, quality class, count-statistical uncertainty, effective sample size and confidence intervals.

Advanced statistics include time profiles, data quality, distribution diagnostics, threshold events, sensitivity analysis, daily tables and heatmaps. Missing hours are not imputed and statistical outliers are not silently removed.

5. Statistical methods

Moving-block bootstrap accounts for dependence between consecutive hourly values. Effective sample size can therefore be much smaller than the stored sample count.

Mann-Kendall and Sen slope describe trends. Skewness, kurtosis, geometric metrics, IQR, MAD and trimmed mean complement the arithmetic mean. Continuous threshold exceedances are analysed as events with duration, maximum, mean and cumulative exceedance area.

The Poisson-based uncertainty estimate covers statistical counting uncertainty only. It excludes calibration uncertainty, systematic device error, placement and representativeness.

Important: original data remains unchanged; calculated values and quality flags are derived information.

6. Expert view

The Expert view exposes firmware, serial path, decoder information, SPIR checksums, raw counts, database state, factor history and runtime diagnostics.

The currently configured factor and the factor stored with a historical measurement are shown separately to preserve reproducibility.

7. Sites, campaigns and events

Sites and sessions are local metadata. Ventilation, window changes, construction, device moves and outages can be annotated and included in analysis and reports.

For scientific work, document room, floor, measurement height, wall distance, ventilation conditions, purpose and responsible person.

8. GQ Radiation World Map

Upload is optional and disabled by default. A persistent queue prevents duplicates, retries temporary errors and applies the configured maximum age and retry limit.

The app sends measurement data and configured GQ identifiers, but no GPS coordinates. Public location data is controlled in the GQ device profile.

9. Reports

Compact and scientific PDF reports include period, site, device, coverage, statistics, threshold analysis, quality information, methods, software/schema versions and a SHA-256 checksum of the selected data.

Reports are documentation aids, not official certificates.

10. Data management

Data management can be disabled completely. It provides backup, export, restore, complete local database reset and complete Home Assistant Recorder purge for RadonScan/Radon Monitoring entities.

Local reset does not erase the physical detector history. Recorder purge may require a long-lived administrator token.

Important: original data remains unchanged; calculated values and quality flags are derived information.

11. Configuration

Key options include polling interval, serial port, conversion factor, minimum coverage, analysis time zone, retention, World Map settings, data-management switch and optional Home Assistant administrator token.

12. Troubleshooting

No device: check USB, permissions, port and competing processes. No new value: only completed hours are imported. Wrong status after upgrade: restart the app and reopen Ingress. Recorder purge fails: verify administrator token and Recorder availability.

13. Privacy and security

Measurements stay local while World Map is disabled. Do not publish credentials in diagnostics or screenshots. Backups may contain measurements, sites, serial identifiers and reports and should be stored securely.

Configuration overview

Option	Purpose
factor_bq_m3_per_cph	Current conversion factor
minimum_data_coverage_percent	Minimum coverage for period metrics
analysis_timezone	Time zone for daily/weekly profiles
data_management_enabled	Show/hide and enable/disable data management
homeassistant_access_token	Optional administrator token for Recorder purge
gmcmmap_enabled	Enable optional World Map area